

UNDERSTANDING GRAMMAR REPRESENTATION IN THE HUMAN BRAIN VIA INFORMATION-RESTRICTED LLMS

SIMONE ESPOSITO

ANUJA NEGI

FATMA DENIZ

KLAUS OBERMAYER

SUPERVISOR

FIRST EXAMINER

SECOND EXAMINER

How does the brain distinguish nouns and verbs?

How does the brain distinguish nouns and verbs? **It's complicated...**

How does the brain distinguish nouns and verbs? **It's complicated...**

Fronto-Temporal Dichotomy

Damasio and Tranel (1993)

Patients with **brain lesions** doing **object** and **action** naming tasks

→ **Temporal** regions involved in the retrieval of **nouns**

→ **Frontal** regions involved in the retrieval of **verbs**

How does the brain distinguish nouns and verbs? **It's complicated...**

Fronto-Temporal Dichotomy

Damasio and Tranel (1993)

Patients with **brain lesions** doing **object** and **action** naming tasks

→ **Temporal** regions involved in the retrieval of **nouns**

→ **Frontal** regions involved in the retrieval of **verbs**

What is driving the distinction?

How does the brain distinguish nouns and verbs? **It's complicated...**

Emergentist View

Bird et al. (2000), Vigliocco et al. (2011)

Noun / Verb distinction is not **lexical** but **semantic**

→ Nouns are **more imageable** and related to **objects**

→ Verbs are **less imageable** and related to **actions**

How does the brain distinguish nouns and verbs? **It's complicated...**

Combinatorial View

Shapiro and Caramazza (2003), Tyler et al. (2003)

Noun / Verb distinction driven by **computational** processes

→ Pseudowords: “*he wugged*” and “*the wugs*” represented in distinct brain areas

→ Verbs are more **computationally demanding**

How does the brain distinguish nouns and verbs? **It's complicated...**

Distributed Account

Crepaldi et al. (2013)

Theory-blind meta-analysis based on 36 neuroimaging studies

→ 32 reliable **clusters**:

3 for **nouns** (R-ITG, L-AG, L-IPG)

1 for **verbs** (R-MTG)

3 both **nouns** and **verbs**; Task dependent (L-IFG, R-IFG, L-Insula)

→ No support for **Fronto-Temporal** dichotomy or **Embodied Cognition**

→ **Distributed** processing of nouns and verbs

Methods

Picture Naming (Shapiro and Caramazza, 2011)

Semantic Categorization (Tyler et al., 2003)

Methods

Picture Naming (Shapiro and Caramazza, 2011)

Semantic Categorization (Tyler et al., 2003)

Passive processing?

Methods

Picture Naming (Shapiro and Caramazza, 2011)

Semantic Categorization (Tyler et al., 2003)

Voxel-wise Encoding Models

Stimuli

- 11 stories from
The Moth Radio Hour
- 10-15 min per story
- Listening and reading

Stimuli

- 11 stories from
The Moth Radio Hour
- 10-15 min per story
- Listening and reading

fMRI Data

- From Deniz et al. (2019)
- 9 subjects

Stimuli

- 11 stories from
The Moth Radio Hour
- 10-15 min per story
- Listening and reading

fMRI Data

- From Deniz et al. (2019)
- 9 subjects

Features

Listening

- Phonemes + count

Reading

- Letters + count
- Word length variation

Both

- Number of words
- Number of pauses
- Lexical (English1000)

Stimulus

There're some scientists who say that the universe that we inhabit is not the only universe there is ...

Stimulus

*There're some scientists who say that the universe
that we inhabit is not the only universe there is ...*

↓ POS tagging via spaCy

Attention Mask

There're	some	scientists	who	say	that	the	universe		
that	we	inhabit	is	not	the	only	universe	there	is

Stimulus

*There're some scientists who say that the universe
that we inhabit is not the only universe there is ...*

↓ POS tagging via spaCy

Attention Mask

*There're some scientists who say that the universe
that we inhabit is not the only universe there is*



GPT-2

Layer 1

⋮

Layer 8

⋮

Layer 12

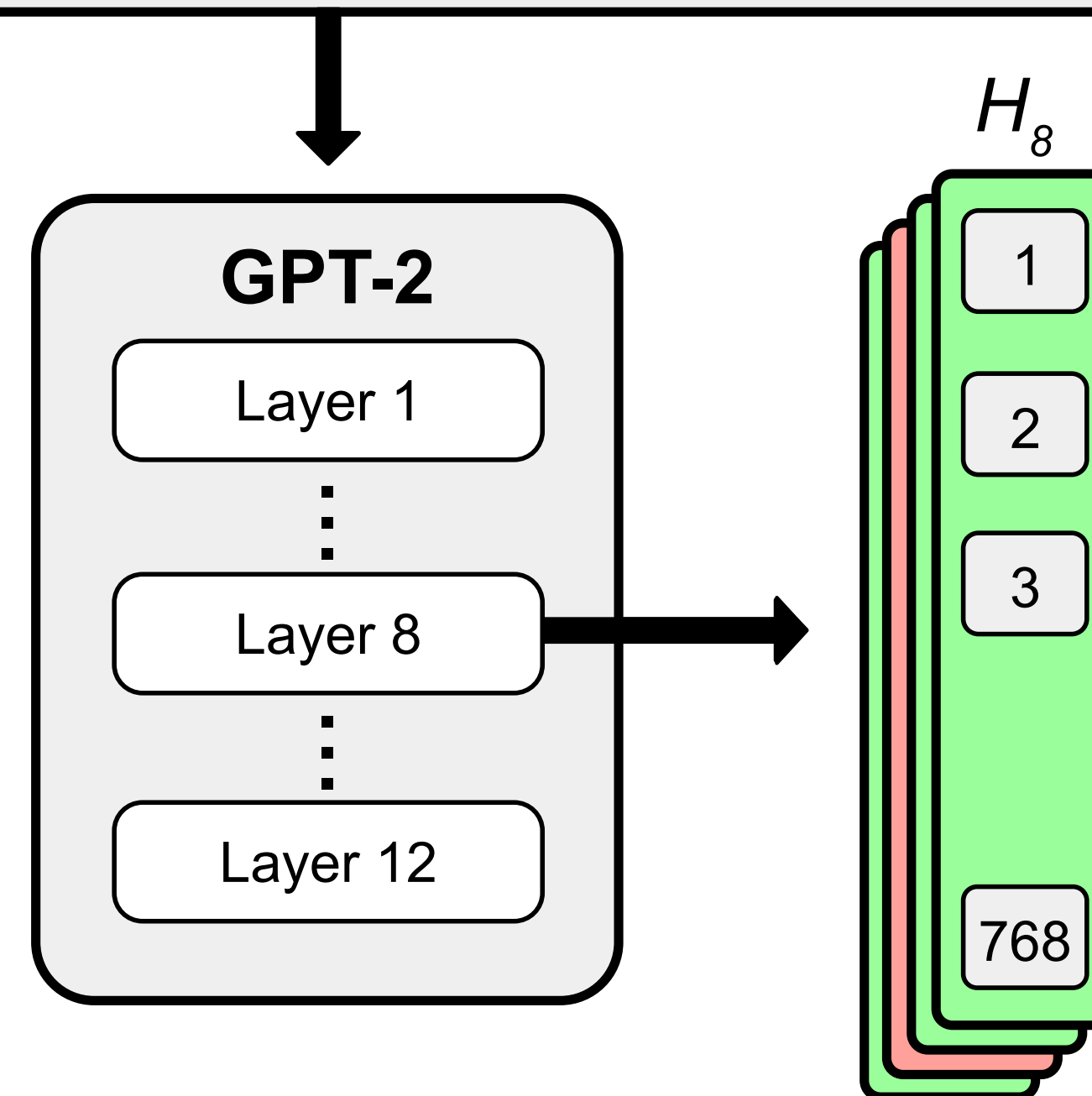
Stimulus

*There're some scientists who say that the universe
that we inhabit is not the only universe there is ...*

↓ POS tagging via spaCy

Attention Mask

*There're some scientists who say that the universe
that we inhabit is not the only universe there is*



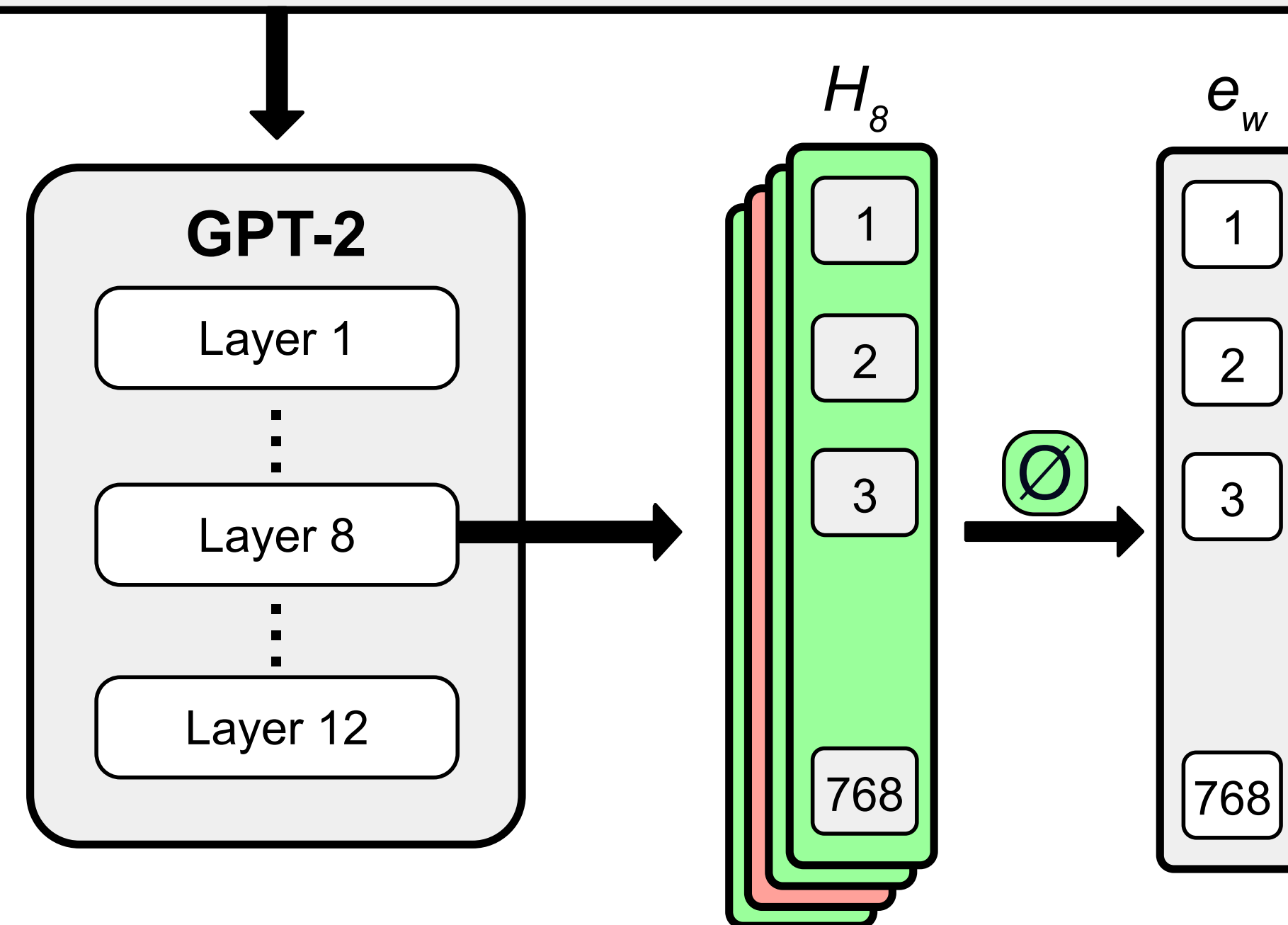
Stimulus

*There're some scientists who say that the universe
that we inhabit is not the only universe there is ...*

↓ POS tagging via spaCy

Attention Mask

*There're some scientists who say that the universe
that we inhabit is not the only universe there is*



Banded Ridge Regression

to model BOLD as linear combination of masked embeddings and low-level features

Banded Ridge Regression

to model BOLD as linear combination of masked embeddings and low-level features

POS used

nouns, verbs, adjectives, adverbs, pronouns, interjections

VOXEL-WISE ENCODING

Banded Ridge Regression

to model BOLD as linear combination of masked embeddings and low-level features

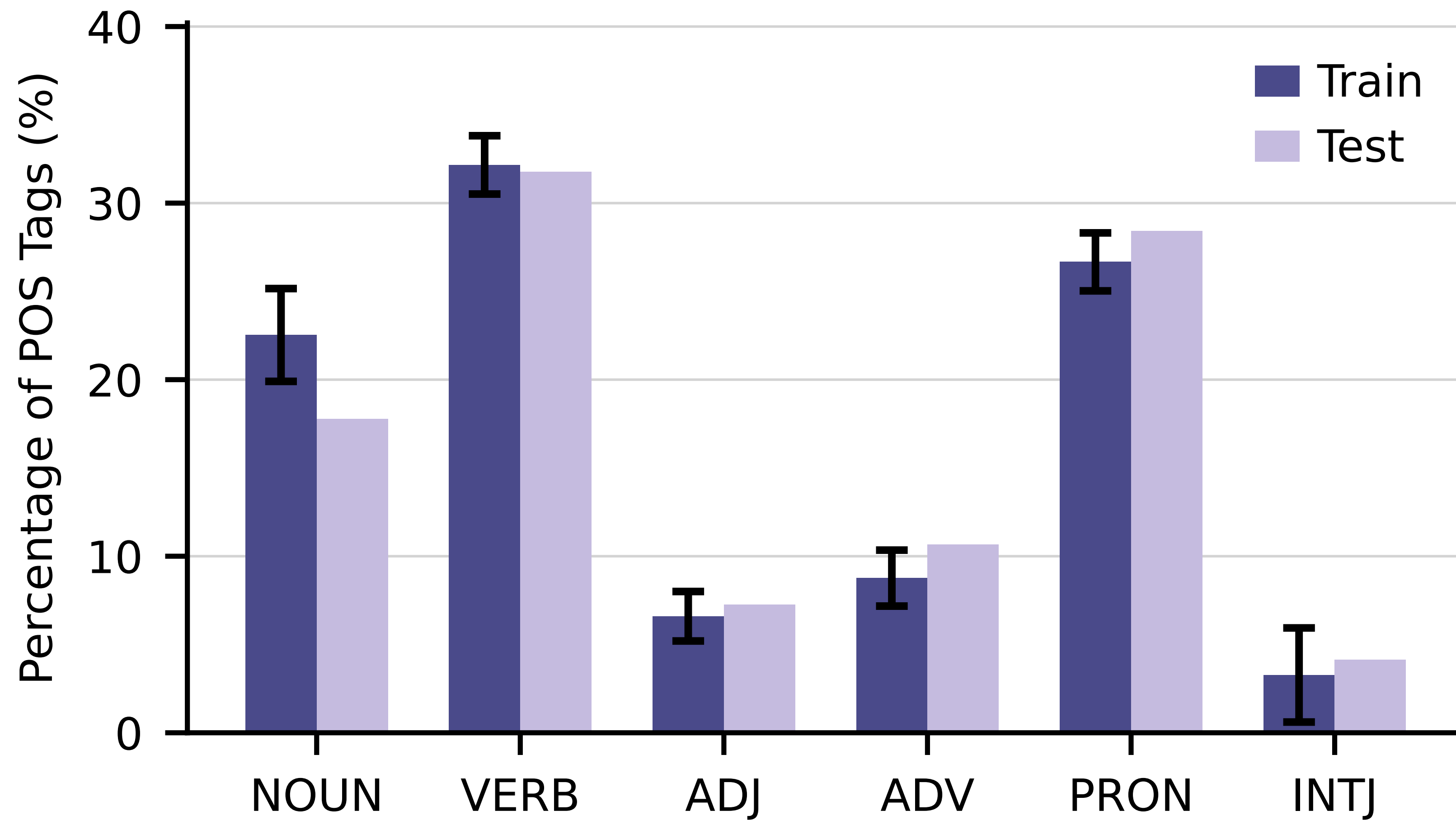
POS used

nouns, verbs, adjectives, adverbs, pronouns, interjections

Prediction accuracy

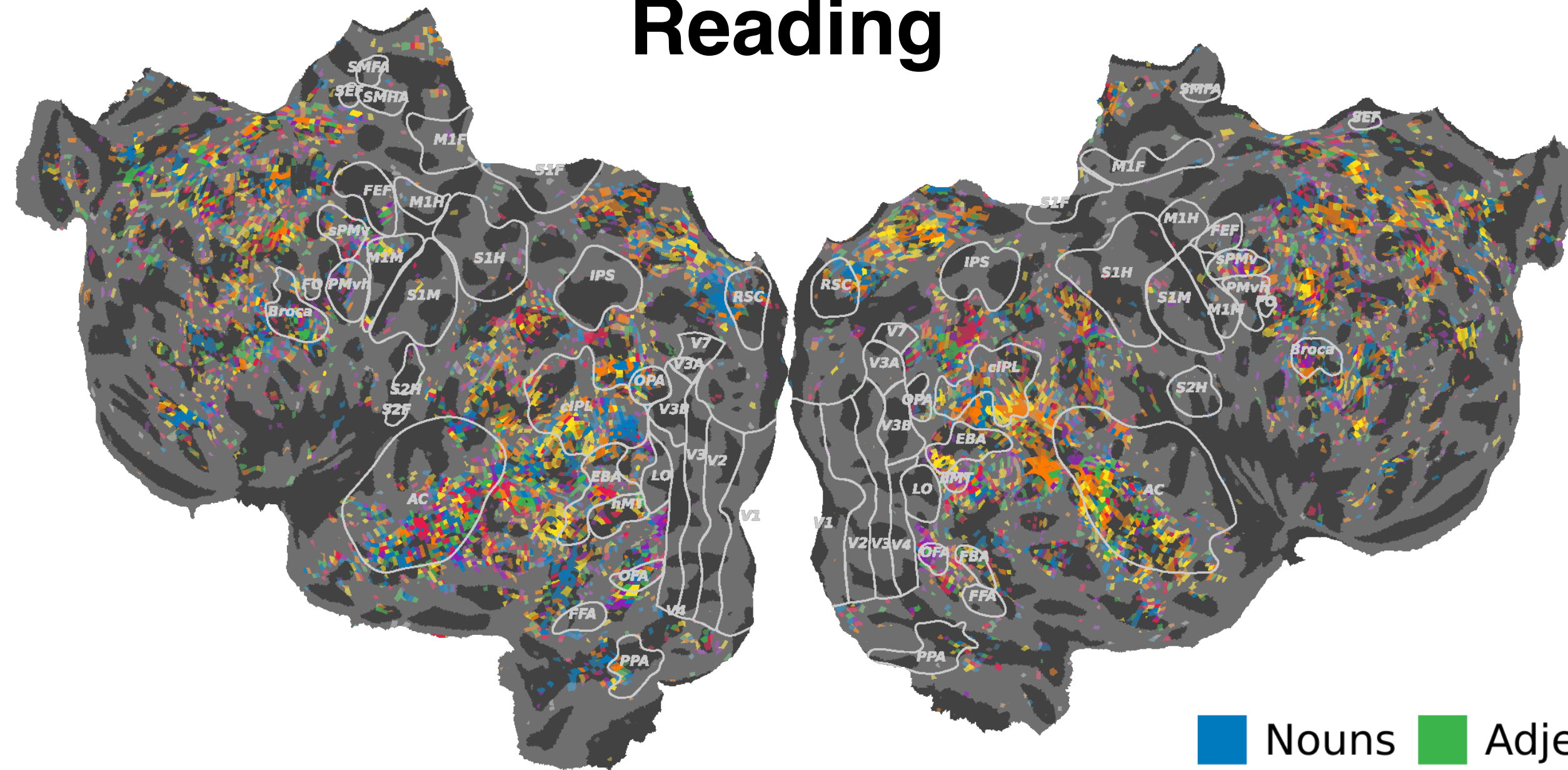
Pearson correlation between predicted and actual train data BOLD response

PARTS OF SPEECH DISTRIBUTION IN DATASET

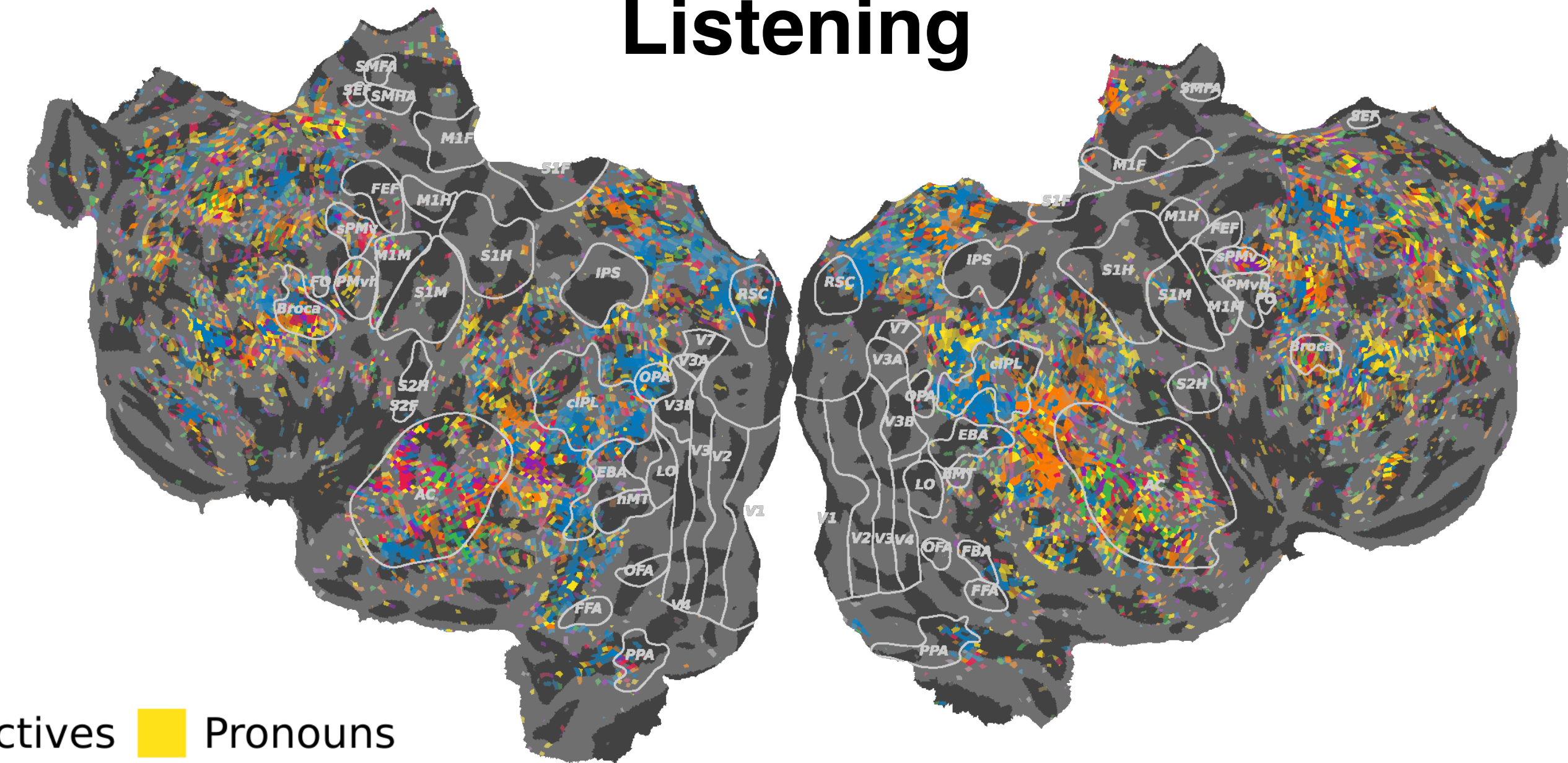


POS-MASKED CONTEXTUAL EMBEDDINGS REVEAL CLUSTERS OF POS SENSITIVITY ACROSS THE BRAIN

Subject 7
Reading



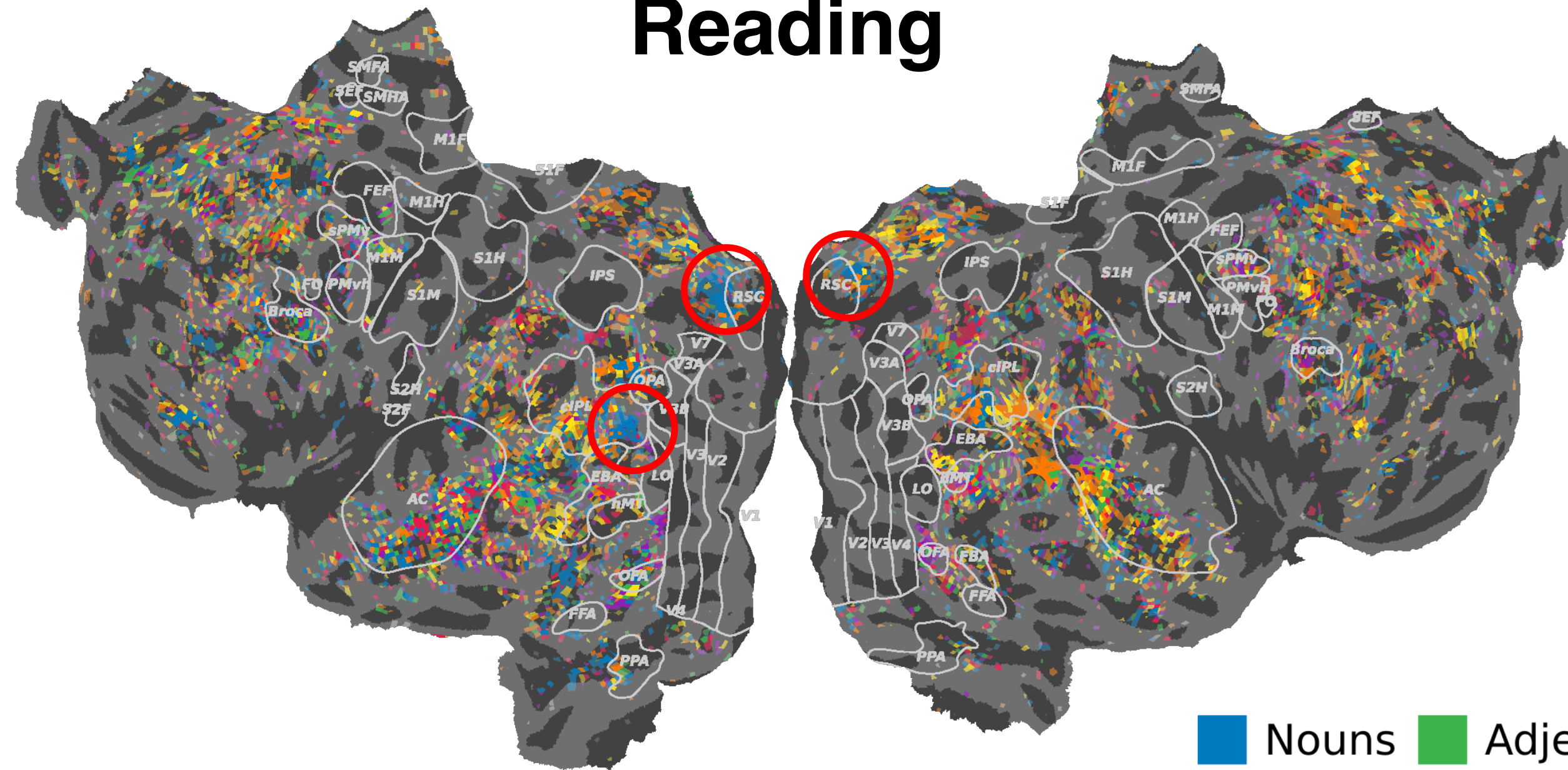
Subject 7
Listening



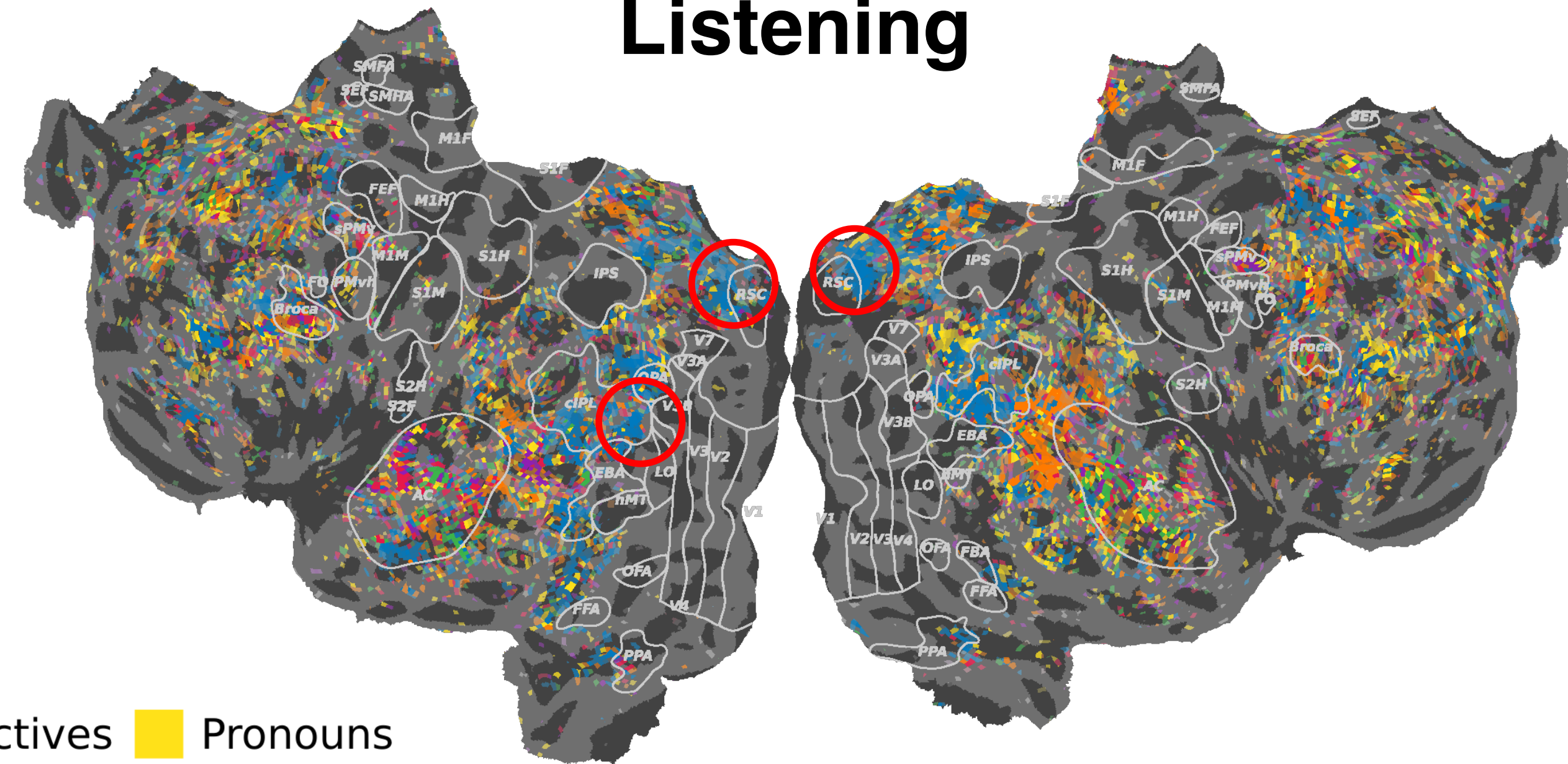
■ Nouns	■ Adjectives	■ Pronouns
■ Verbs	■ Adverbs	■ Interjections

POS-MASKED CONTEXTUAL EMBEDDINGS REVEAL CLUSTERS OF POS SENSITIVITY ACROSS THE BRAIN

Subject 7
Reading



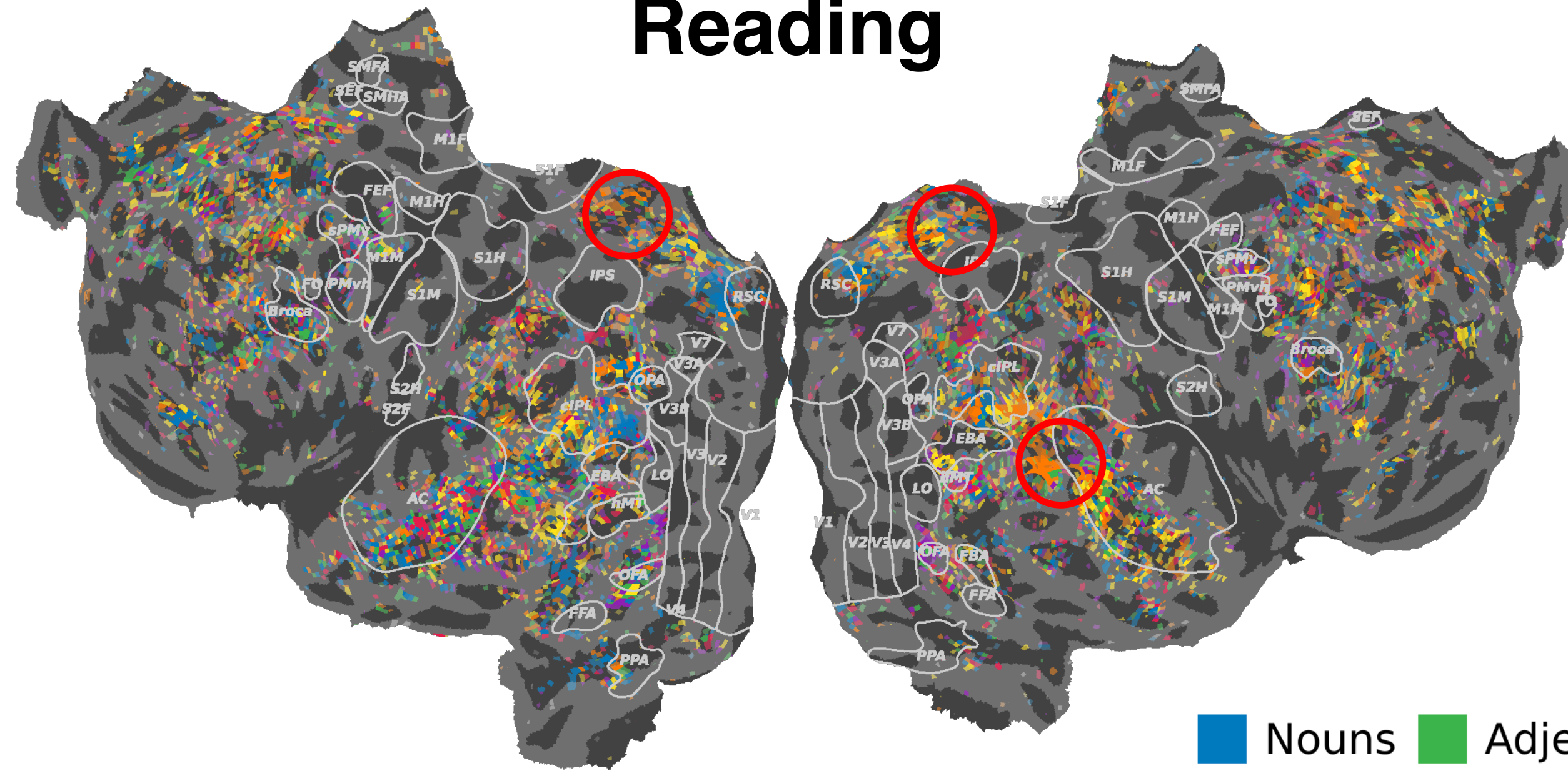
Subject 7
Listening



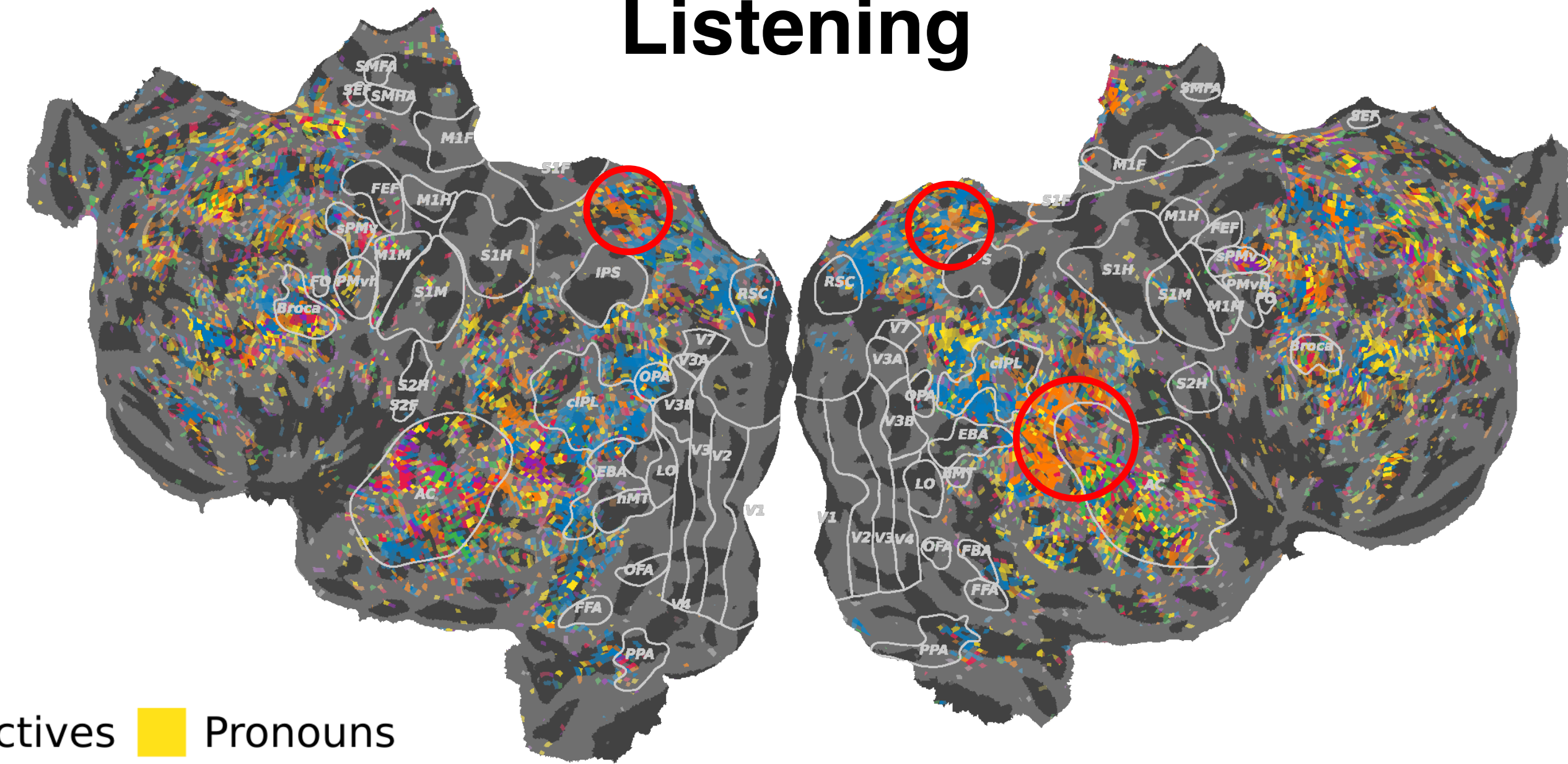
■ Nouns ■ Adjectives ■ Pronouns
■ Verbs ■ Adverbs ■ Interjections

POS-MASKED CONTEXTUAL EMBEDDINGS REVEAL CLUSTERS OF POS SENSITIVITY ACROSS THE BRAIN

Subject 7
Reading



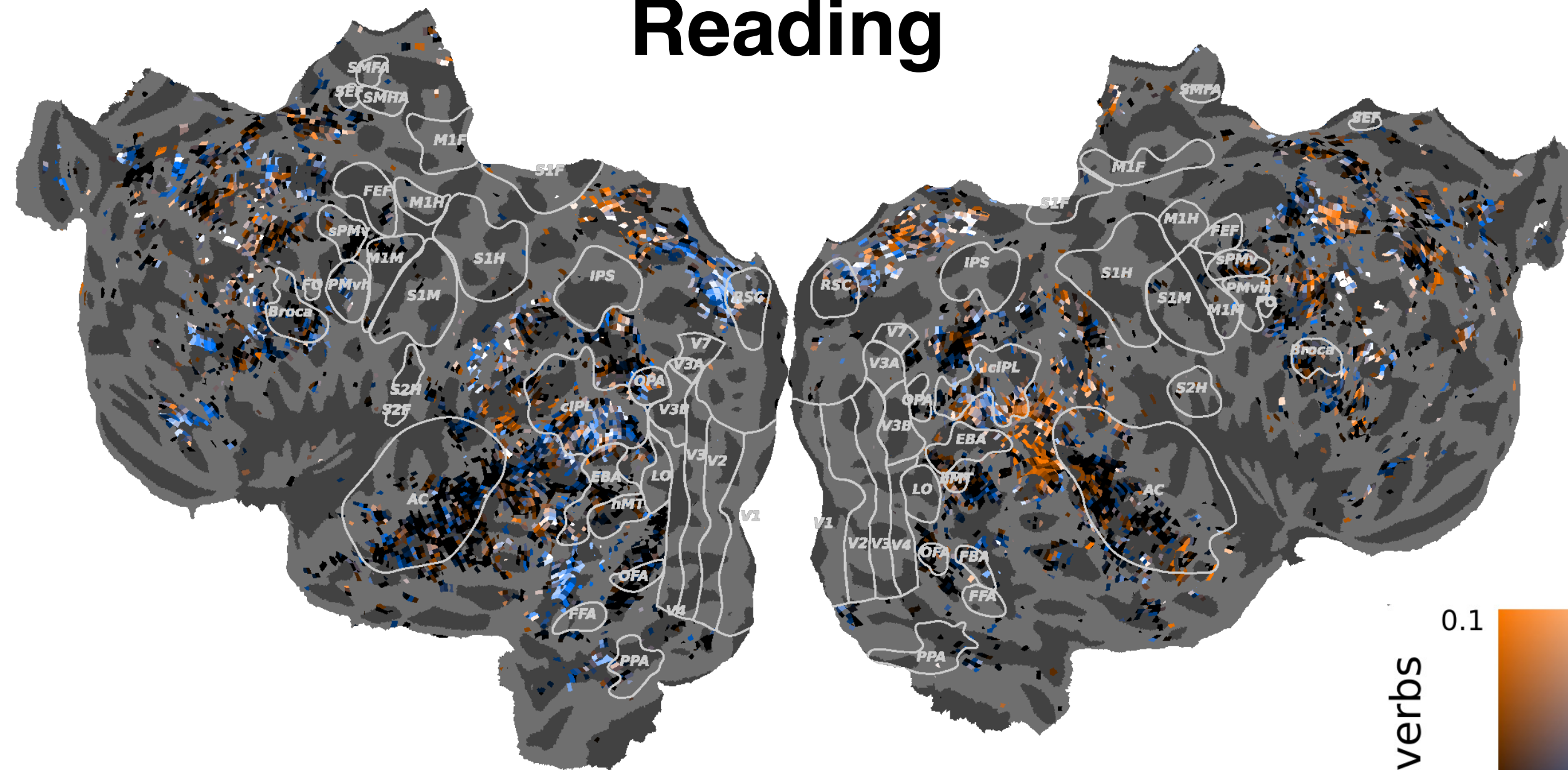
Subject 7
Listening



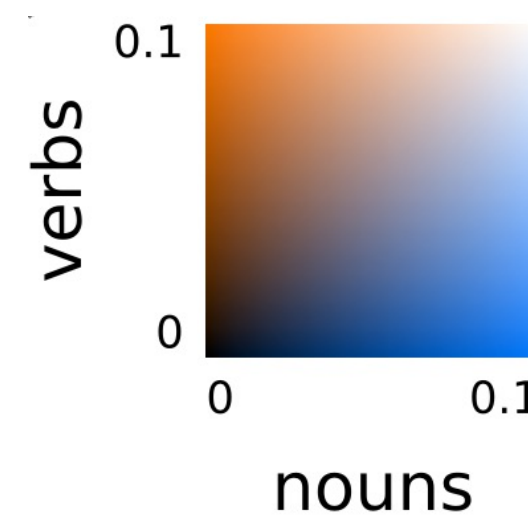
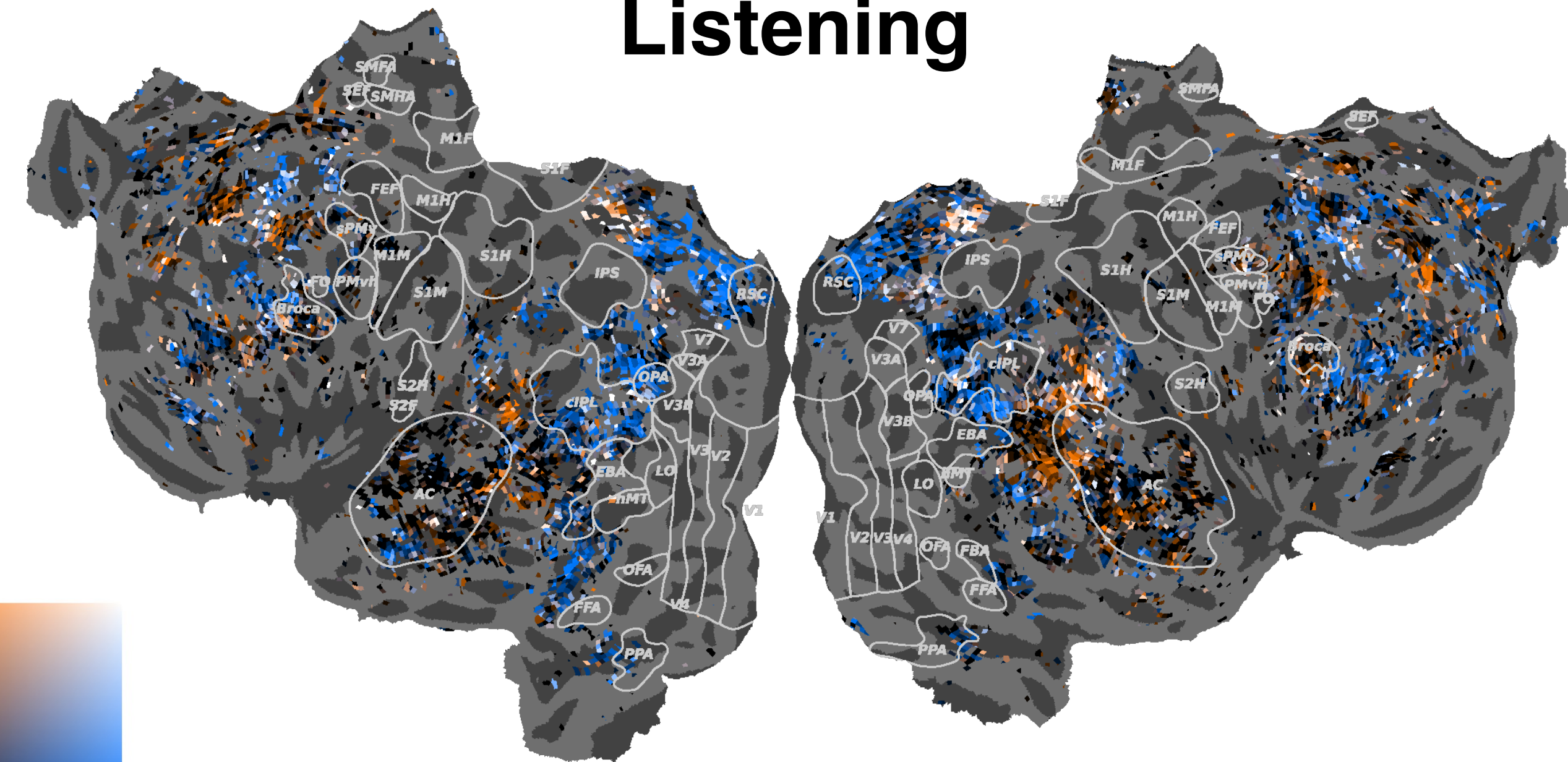
■ Nouns ■ Adjectives ■ Pronouns
■ Verbs ■ Adverbs ■ Interjections

NOUNS AND VERBS SHOW DISTINCT PATTERNS OF SENSITIVITY ACROSS THE BRAIN

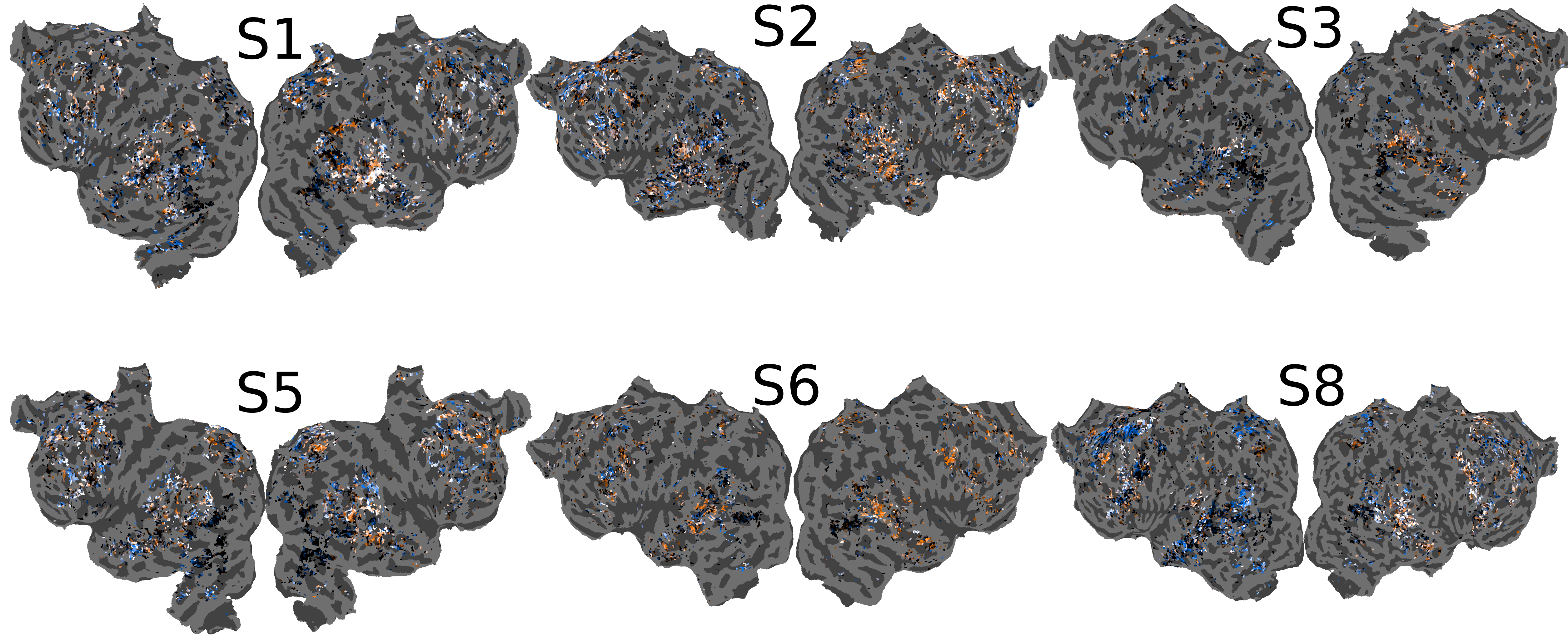
Subject 7
Reading



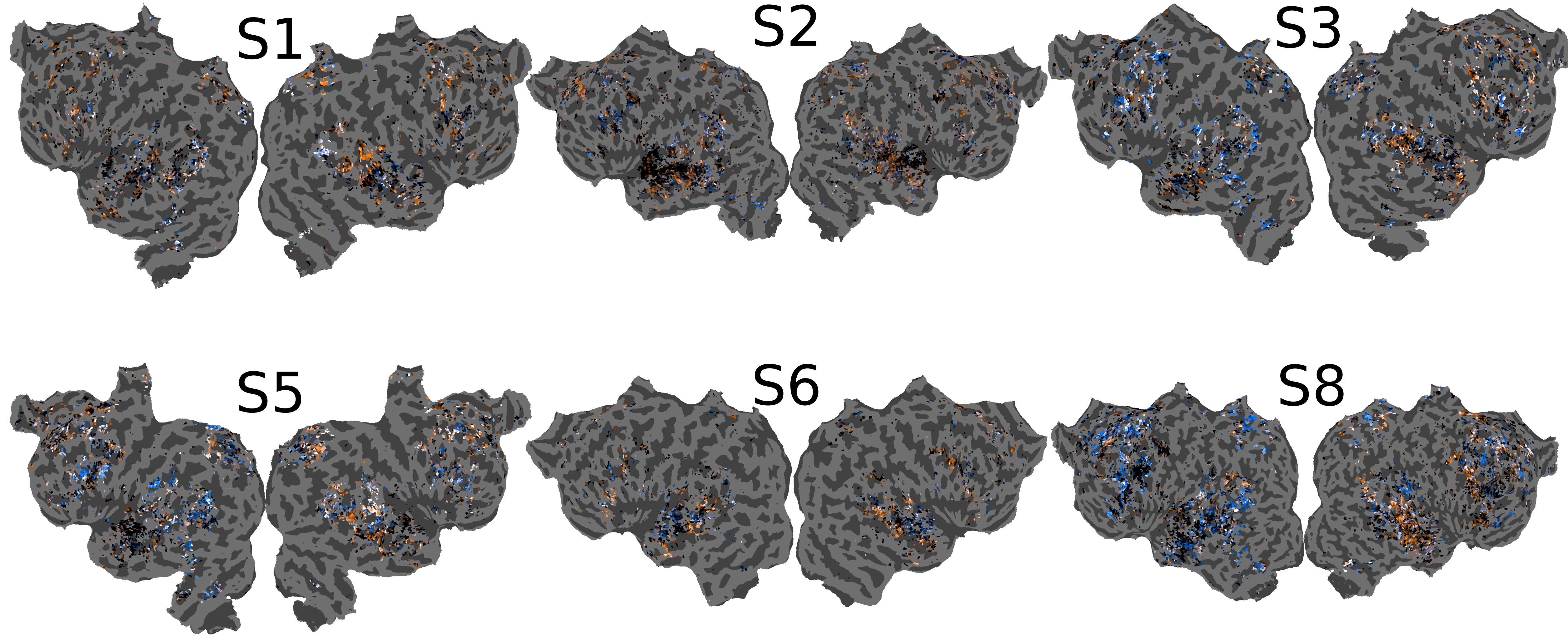
Subject 7
Listening

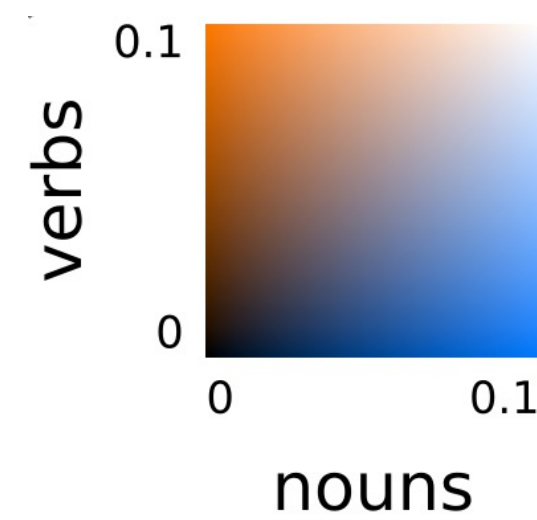


NOUN AND VERB SENSITIVITY – READING



NOUN AND VERB SENSITIVITY – LISTENING

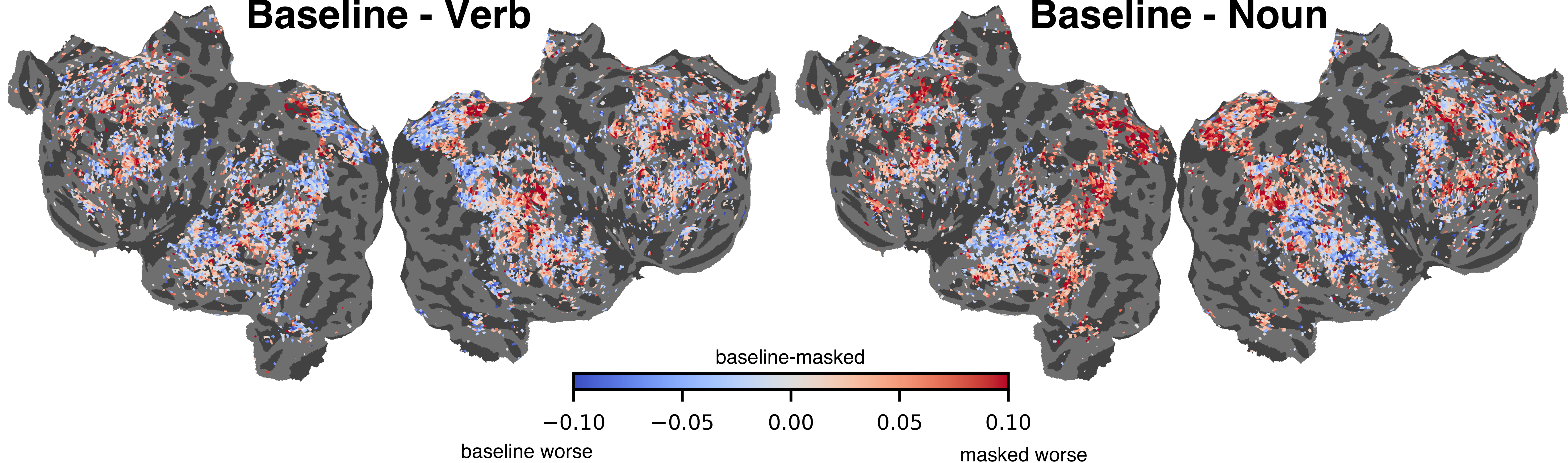




MASKING NOUNS AND VERBS IMPROVES PREDICTION ACCURACY IN SOME AREAS

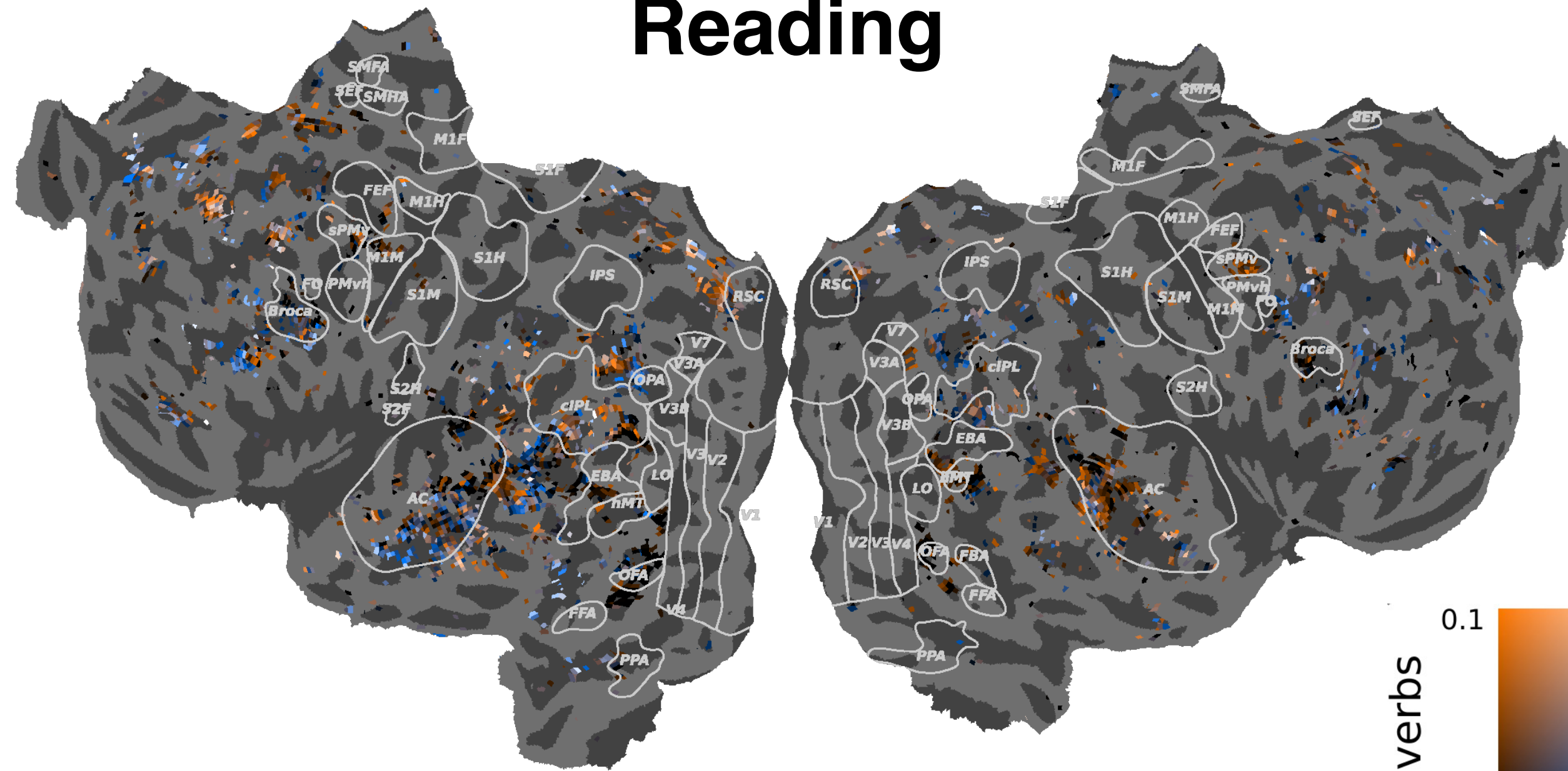
Subject 7
Baseline - Verb

Subject 7
Baseline - Noun

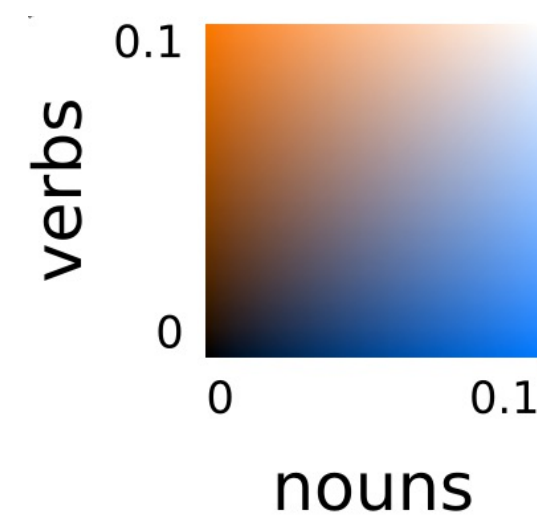
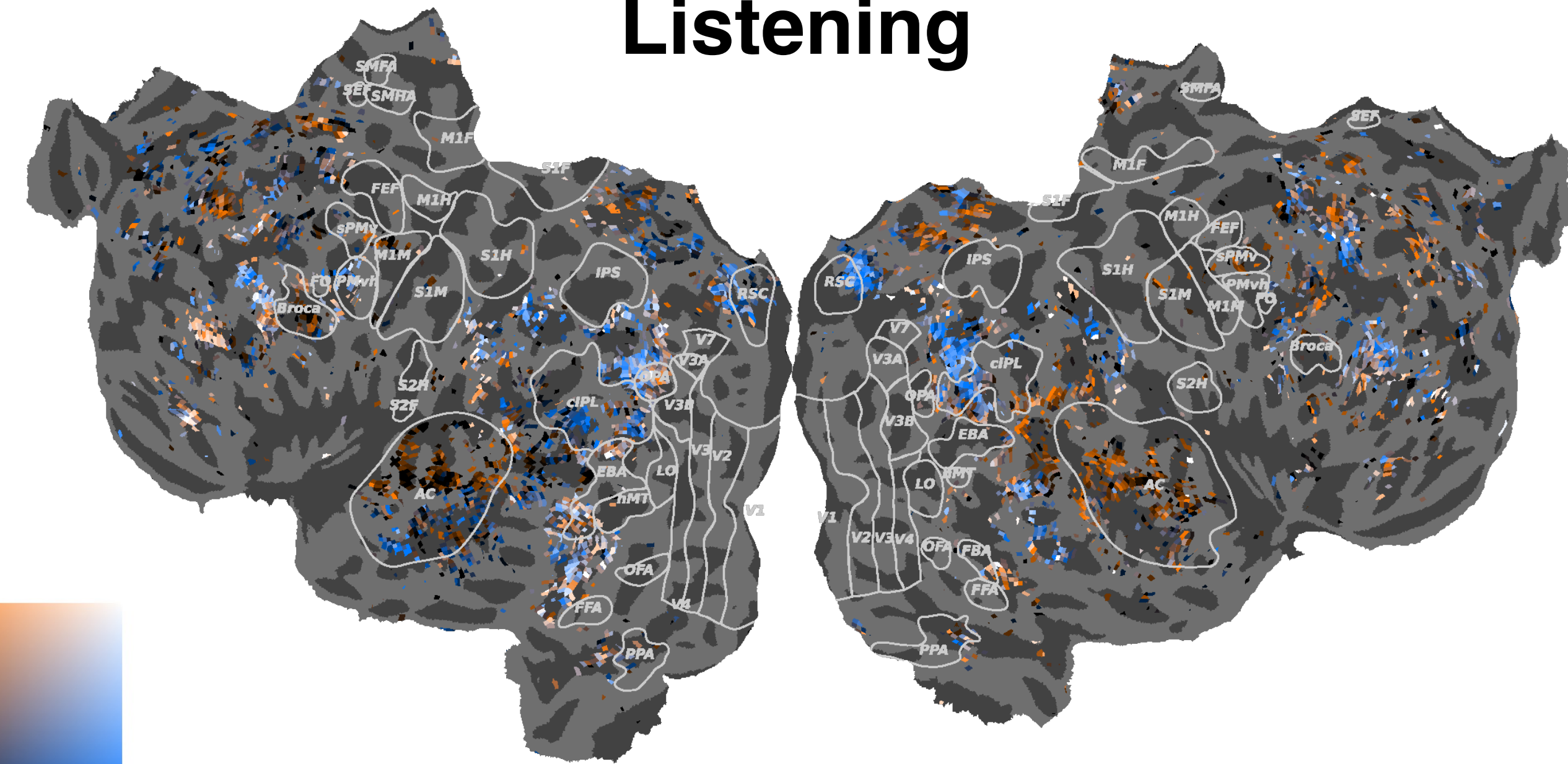


MASKING OF LEXICAL EMBEDDINGS PRODUCES SIMILAR SENSITIVITY MAPS

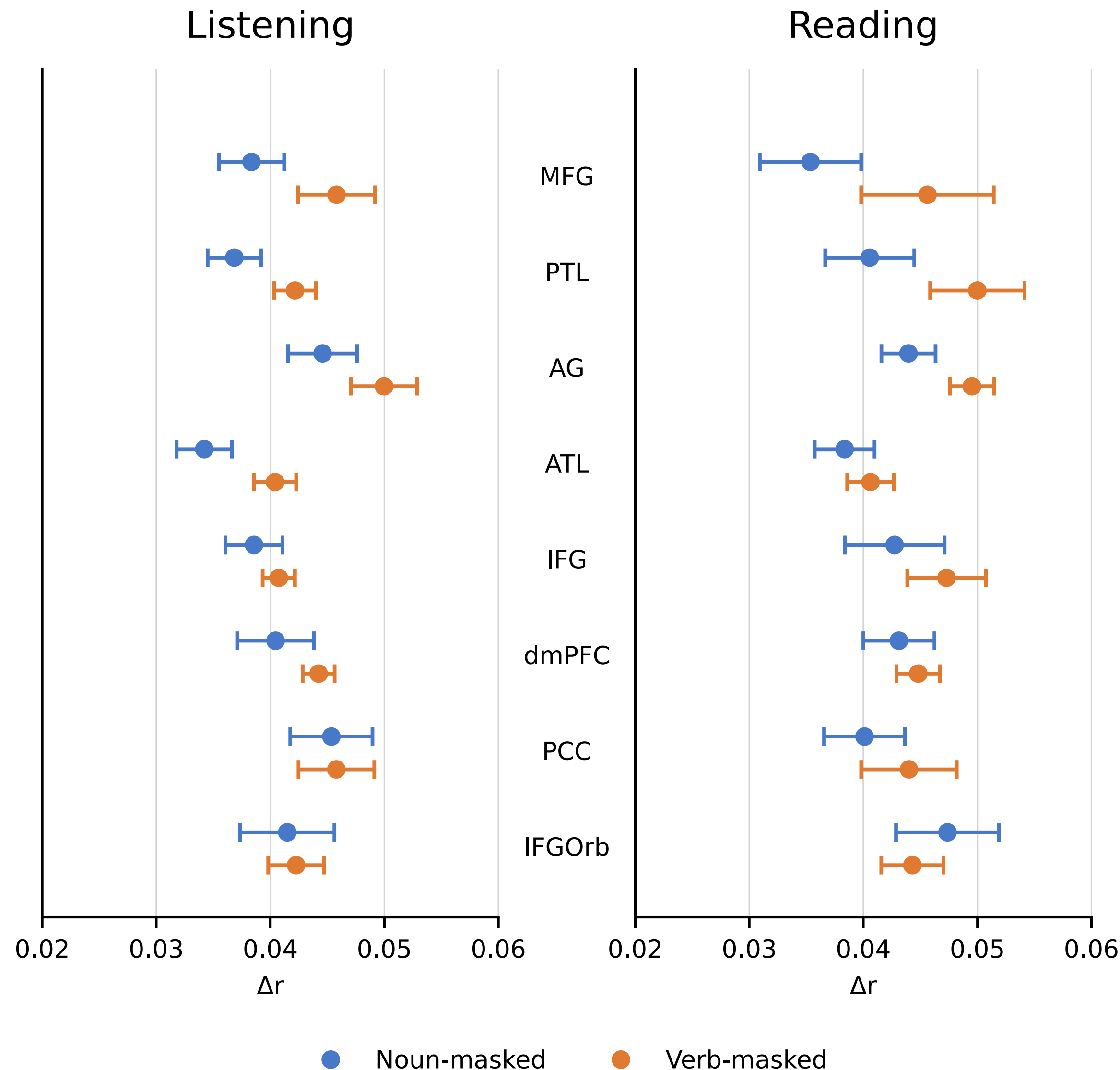
Subject 7
Reading



Subject 7
Listening



VERB SENSITIVITY IS MORE PREVALENT ACROSS LANGUAGE ROIS



Language ROIs

Fedorenko et al. (2010)

Mean Drop

0.03 – 0.06

Verb > Noun

MFG, PTL, AG, ATL

Reading vs Listening

Similar across ROIs

COMPARISON WITH LITERATURE

Theoretical account	Predicted pattern	Observed pattern
Fronto-Temporal View <i>Neuropsychological</i> (A. R. Damasio and Tranel 1993)	V > N in frontal regions; N > V in temporal regions	V > N in temporal regions (ATL, PTL); no clean frontal/temporal split.
Emergentist View <i>Object-Action, Imageability</i> (Bird et al. 2000; Vigliocco et al. 2011)	V > N in frontal and parietal regions; N > V in inferior temporal regions.	N > V near visual areas; Similar sensitivity in frontal regions.
Combinatorial <i>Morphology/inflection</i> (Tyler et al. 2004; K. A. Shapiro et al. 2005)	N = V broadly, with V > N specifically in IFG for morphological composition.	Slight V > N in IFG; N > V in IFGOrb.
Distributed <i>Shared network</i> (Crepaldi et al. 2013; Caucheteux et al. 2021)	Noun and verb circuits distributed across frontal, parietal, and temporal regions.	Smaller clusters of noun and verb sensitive voxels across the brain.

Noun and verb sensitivity forms **distinct** but **distributed clusters** across frontal, parietal and temporal cortex

Noun and verb sensitivity forms **distinct** but **distributed clusters** across frontal, parietal and temporal cortex

Noun and verb processing is **consistent across stimulus modality**

Noun and verb sensitivity forms **distinct** but **distributed clusters** across frontal, parietal and temporal cortex

Noun and verb processing is **consistent across stimulus modality**

Verb sensitivity dominates across most language ROIs

CONCLUSIONS

Noun and verb sensitivity forms **distinct** but **distributed clusters** across frontal, parietal and temporal cortex

Noun and verb processing is **consistent across stimulus modality**

Verb sensitivity dominates across most language ROIs

Results support a **distributed account**

LIMITATIONS

Only the **absence** of nouns / verbs is being investigated

LIMITATIONS

Only the **absence** of nouns / verbs is being investigated

ROIs used in quantification don't cover all clusters /
Clusters split across multiple ROIs

LIMITATIONS

Only the **absence** of nouns / verbs is being investigated

ROIs used in quantification don't cover all clusters /
Clusters split across multiple ROIs

Cross-Modality **not confirmed**

Investigating *presence* of nouns and verbs

Using one-hot encodings

Investigating *presence* of nouns and verbs

Using one-hot encodings

Using a different dataset

Other than Moth

Investigating *presence* of nouns and verbs

Using one-hot encodings

Using a different dataset

Other than Moth

Looking at a different Language

Other than English, e.g. Chinese

Investigating *presence* of nouns and verbs

Using one-hot encodings

Using a different dataset

Other than Moth

Looking at a different Language

Other than English, e.g. Chinese

Cluster Categorization

Which kind of nouns / verbs in the clusters?